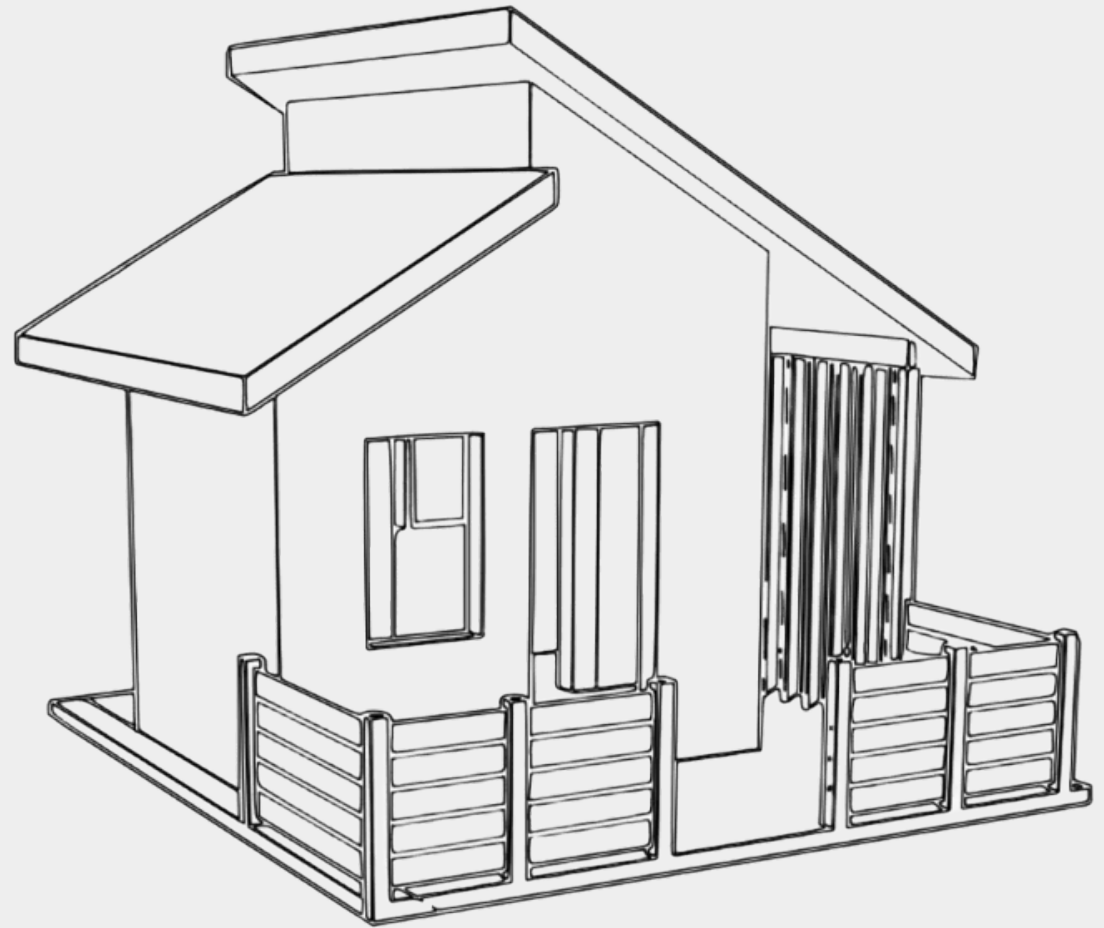


The Tiny Retreat Project

The Duo Den

THE DESIGN NARRATIVE



Problem Statement

1. Low exposure to greenery
2. Lack of lighting
3. Noise level
4. Temperature
5. Layout
6. Odour



Precedents



(Santonori Photos, n.d.)



(Meloni, n.d.)



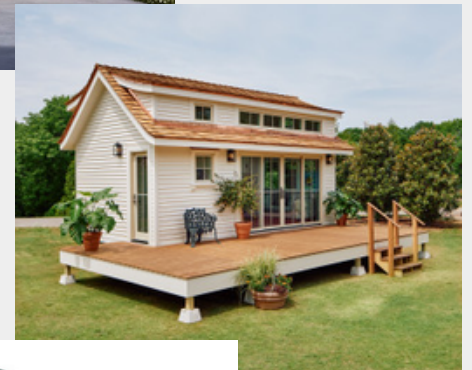
(Tiny House Loft, 2024)



(HGTV, 2021)



(Sunshine Home Designs, 2021)

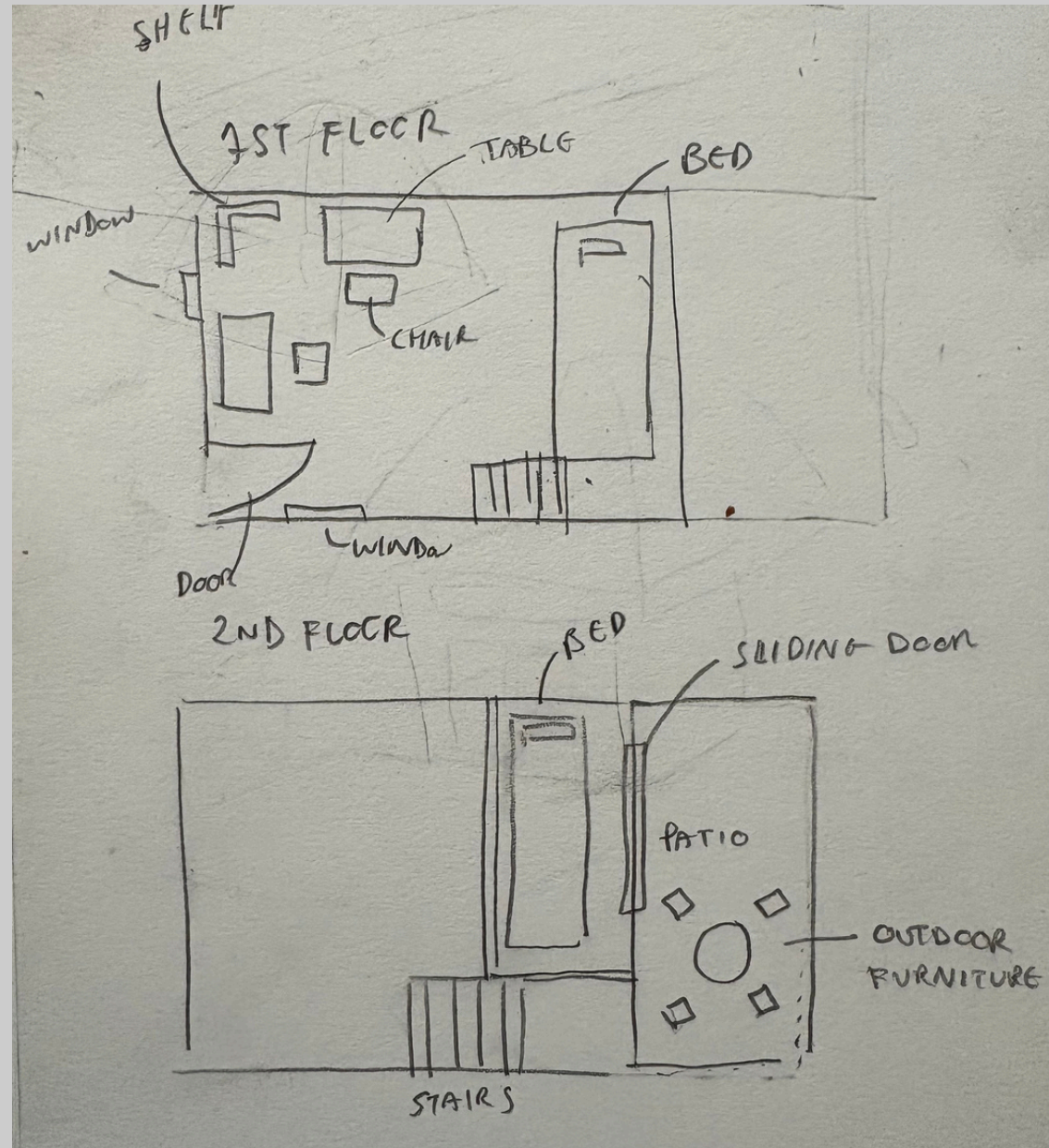
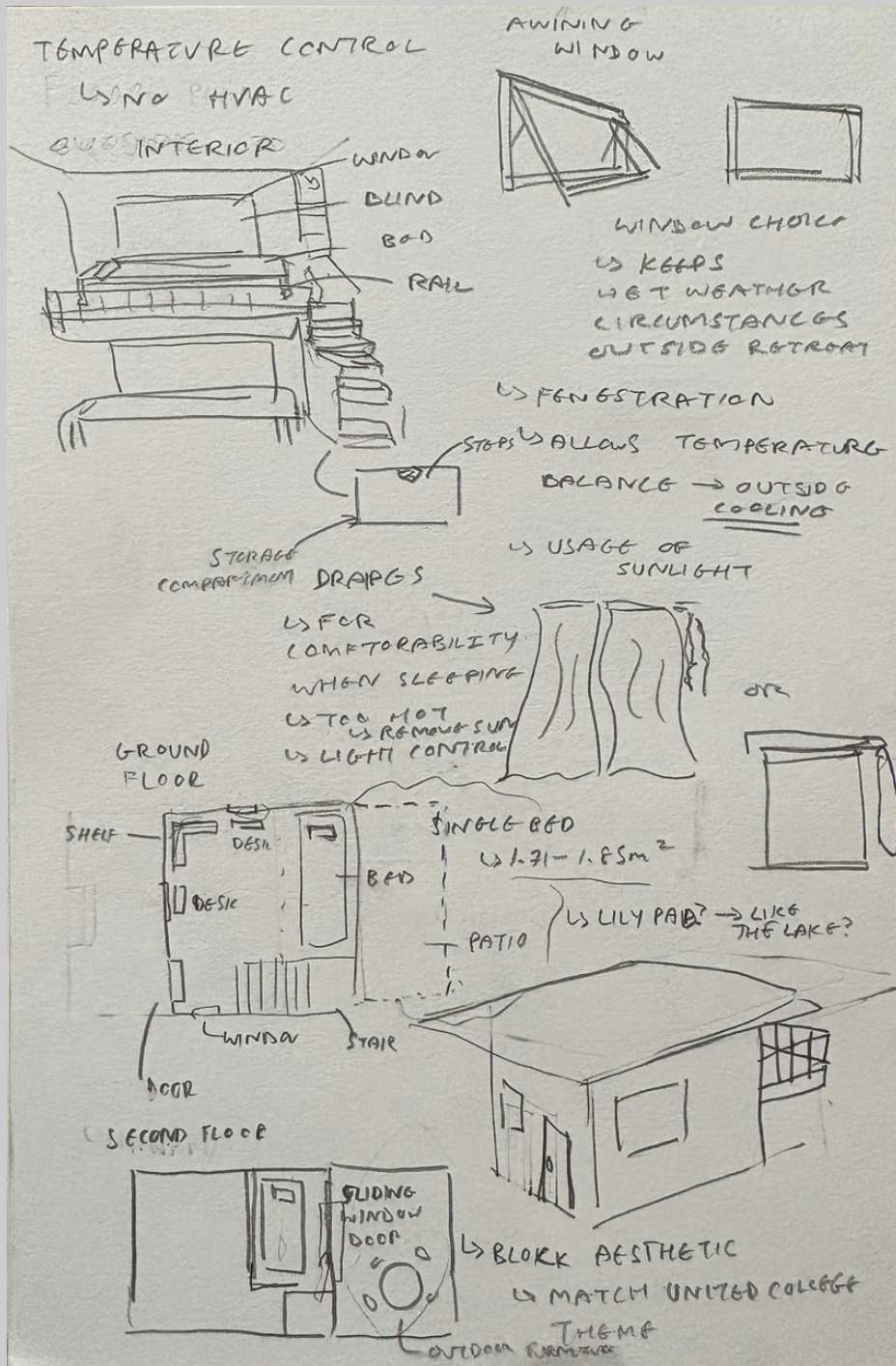


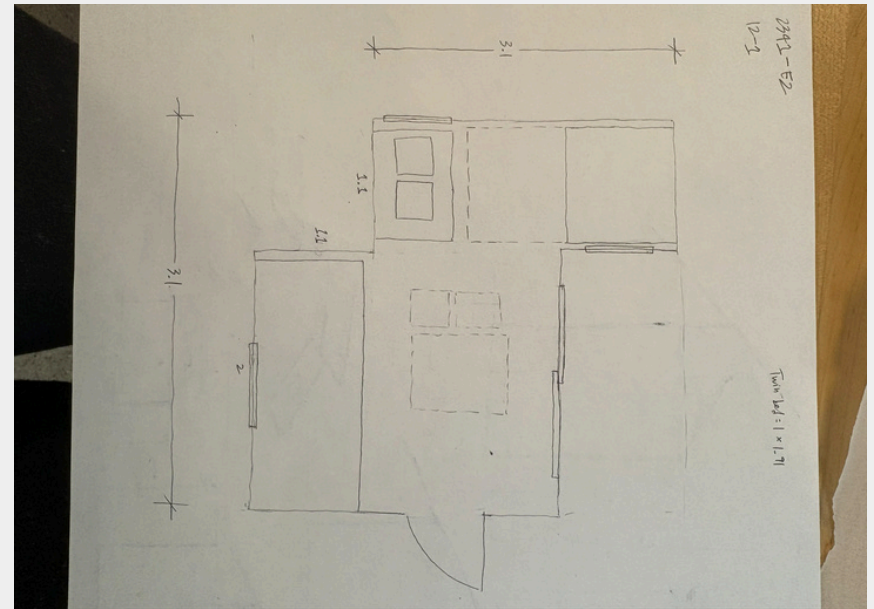
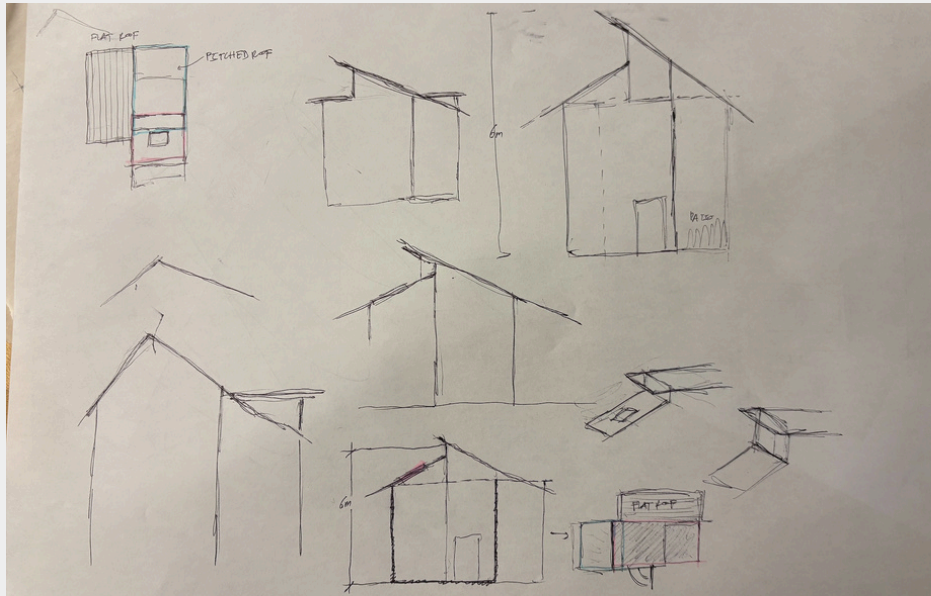
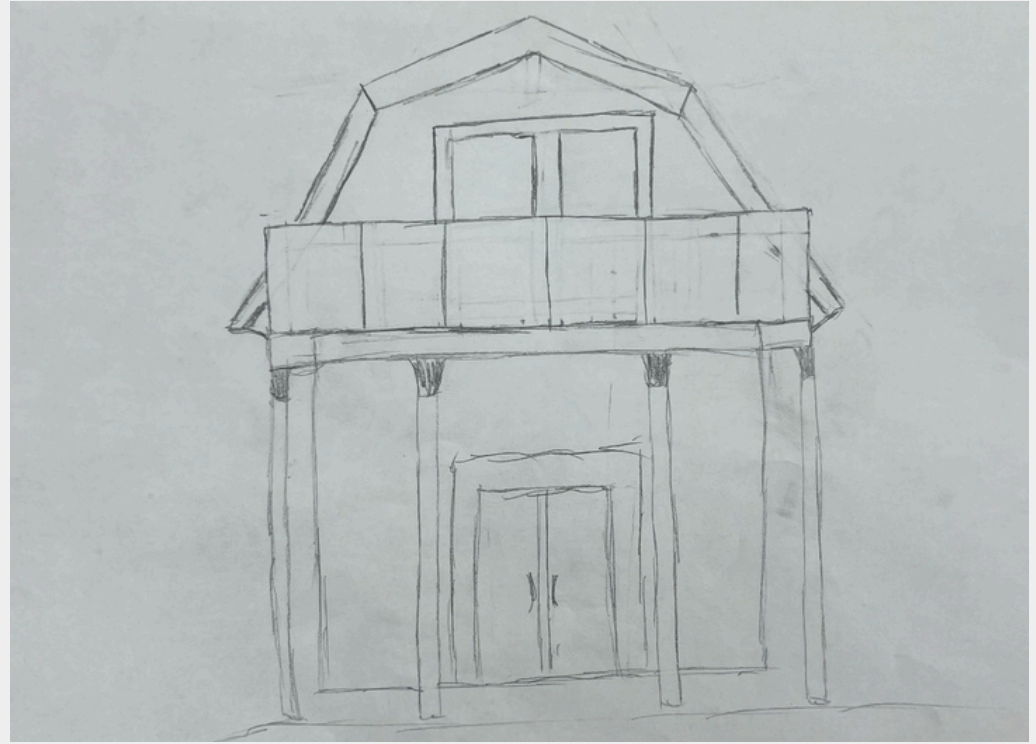
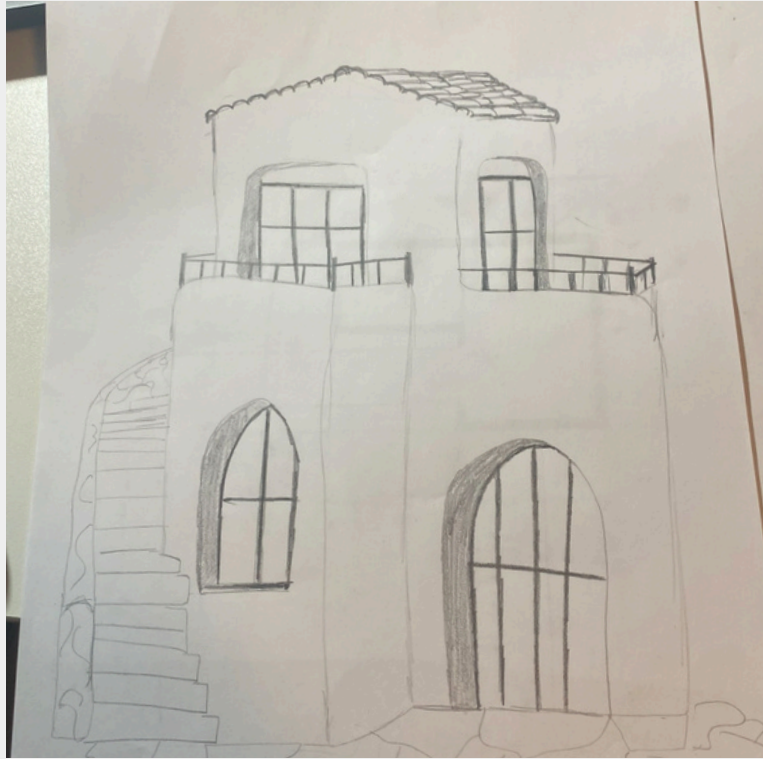
(Toplu, 2022)



(Metal Siding gallery, 2020)

Concept Designs





The retreat location was chosen to be on a flatter plane for easy access from the path as well as the preexisting path which leads to washroom usage in the building below.



Site Documentation



Research Report Conclusions

Wall Research:

- Research was conducted for three separate wall systems.
- Fibre cement panelling or aluminum paneling (A2-s1)
- Necessary vented air gap
- Drained and vented housewrap or fluid membrane
- Plywood sheathing
- Graphite EPS & Rockwool Comfortbatt, or rigid insulation board & graphite EPS
- Smart Vapour Retarder
- drywall and interior paint finish

Roof Research:

- Skillion and Lean-to Roof
 - promotes natural ventilation and lighting through high wall
 - efficient water drainage due to consistent slope
- Snap-lock standing seam system
 - performs best on moderate slopes
 - long lasting

Fenestration Research:

- Casement window
 - maximizes air flow and daylight
- Double-glazed window
 - thermal insulation: slows heat transfer
 - sound insulation
- Low emissivity coating
 - thermal control (reflects heat) and energy efficient
- Argon filled gas
 - Better insulation: argon is more dense than air

Foundation Research:

- Compared spread footing slab and monolithic slab on grade
- Spread footing is the standard for most residential buildings
- Monolithic slab on grade can work if properly insulated
- Interior section: Continuous spread footing
- Exterior section: Monolithic slab on grade

References

Cheng, Nina. 2025. "How Background Noise Affects Study Efficiency." Blissful Balances.
<https://blissfulbalances.com/articles/background-noise-study-efficiency/>.

Joye. 2025. "Heater Guides." How Temperature Changes Impact Insulation Performance.
<https://insulation.org/io/articles/insulation-and-temperature-a-useful-relationship/>.

McDaniel, Tina, and Joel McRae. 2024. "The Role of Windows in Home Design: Connecting Your Home to the Outdoors." Harry Norman. <https://www.harrynorman.com/posts/the-importance-and-beauty-of-windows-in-connecting-with-nature/>.

Meloni, Claudia. n.d. "Exploring Dream Homes." Pinterest. <https://ca.pinterest.com/pin/633670610109672323/>.

"Metal Siding Gallery." 2020. Ideal Siding. <https://idealsiding.com/gallery/metal/>.

"Santorini Photos - Download Free High-Quality Pictures." n.d. Freepik. <https://www.freepik.com/photos/santorini>.

"Sunshine Home Design | Single Storey." 2021. Nexa Homes. <https://www.bing.com/images/search?view=detailV2&ccid=V7WythD%2B&id=DA9C896EA59B7124909A176F338C1F2F23EAB031&thid=OIP.V7WythD-Dpoe4hVRijlvowHaEZ&mediaurl=https%3A%2F%2Fbrolenhomes.com.au%2Fstorage%2Fapp%2Fwebp%2Ffad%2F4a9%2F9e1%2F1417x843-facade-lifestyle->

"Tiny House Loft." 2024. AR.InspiredPencil. <https://ar.inspiredpencil.com/pictures-2023/tiny-house-loft>.

Toplu, Ali. 2022. "Cozy Tiny House with High Ceilings." Living in a Tiny. <https://livinginatiny.com/cozy-tiny-house-with-high-ceilings/>.

"Tour a Tiny Barn Home That's Big on Style and Design." 2021. HGTV. <https://www.bing.com/images/search?view=detailV2&ccid=wi7W280X&id=E5AD114249165A9BCB5E43DAFA0AB028624B300C&thid=OIP.wi7W280XRdwW2qcRv7n5hwHaJ5&mediaurl=https%3A%2F%2Fhgtvhome.sndimg.com%2Fcontent%2Fdam%2Fimages%2Fhgtv%2Ffullset%2F2021%2F6%2F29%2F2%2FDOTY20>.